SUBSTANTIVE BASES FOR RECOVERY FOR INJURIES SUSTAINED BY PRIVATE INDIVIDUALS AS A RESULT OF FALLEN SPACE OBJECTS +

Lawrence P. Wilkins *

Space vehicles and their fragments have been falling back to earth since we began putting them up. So far no serious injury has been reported. The recent Cosmos incident sparked interest in the problem of whether, in the event a space object does cause harm in its return to the surface, there are adequate regimes for recovery for that harm.

My treatment of the problem will focus upon three different space objects: one owned by the hypothetical nation of Norka; one owned by the United States Government; and one owned by a private U.S. corporation. The person harmed will be presumed to be a U.S. national. It will be further presumed that Norka, like the U.S., is party to the Outer Space Treaty and the Convention on International Liability for Damage Caused by Space Objects (hereinafter Liability Convention).[1]

Liability of Norka:

Our U.S. national (for convenience, we'll call him Nat) injured by the space object of Norka can benefit from the provisions of the Liability Convention if he can establish that:

(1) he was damaged;

(2) the instrumentality was a space object;

(3) the damage was caused by the space object;

(4) Norka launched, procured the launch, or was the state from whose territory or facility the space object was launched.

Once Nat has established these prerequisites, Norka is absolutely liable without limitation on amount for his injuries.

To clear the first of these hurdles, Nat (or his representative) must show that he has sustained". . . loss of life, personal injury or other impairment of health; or loss of or

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* Assistant Professor of Law, University of Akron School of Law.

[1] Convention on International Liability for Damage Caused by Space Objects, March 29, 1972 (1973) 24 U.S.T. 2389, T.I.A.S. 7762 (hereinafter cited as Liability Convention).

damage to property. . ."[2] Professor Alexander has focused his attention upon the ramifications of "damage" and what it may mean in varying contexts, so I will resist the temptation to launch my own foray of injury. Suffice it to say here, that Nat will recover for the injuries he suffers, so long as those injuries are incurred to the interests in his bodily integrity and property identified in the definition of "damage."

But how does he *establish* his damage? The Preamble to the Liability Convention enunciates and purports to satisfy ". . . the need to elaborate effective international rules and procedures concerning liability for damage caused by space objects. . .," and yet, inspection of the body of the document reveals a gap in those rules and procedures. Article IX instructs Nat that his government will submit a "claim for compensation" to Norka through diplomatic channels, and Article XIV tells him that if the diplomatic negotiations fail, a Claims Commission will be established so that the merits of the claim can be decided. But nowhere does the Liability Convention elaborate upon what constitutes a claim, what items of proof are to be required, and how the allegations are to be evaluated in order to determine if "damage" has indeed occurred. *Compensation,* it is said in Article XII, is to ". . . be determined in accordance with international law and principles of justice and equity. . .". But there is only silence as to how the merits of the assertion by Nat that he has suffered *damage* are to be determined.

Similar problems exist with respect to the items that pertain to the merits of Nat's claim that an instrumentality of Norka was a space object and that the space object caused Nat's damage. The Liability Convention provides only that ". . .'space object' includes component parts of a space object as well as its launch vehicle and parts thereof." Perhaps "space object" has a sufficiently understood meaning common to international states which can be further clarified upon demonstration of need by reference to the draft definitions considered by the Legal Subcommittee, but it is curious that the opportunity to elaborate upon that definition to avoid some obvious problems was passed up by that body. For example, whether articles carried by space vehicles but not specifically designed to move in outer space are to be subsumed in the term was not made clear in the final document.[3]

How strict will the requirements of proof be that the instrumentality was a space object? In the event of complete destruction of the instrumentality upon impact, will it be presumed in favor of Nat that it was indeed a space object, or will the presumption run in favor of Norka? It would seem essential that these kinds of rules be elaborated in such an important document that was so long in coming.

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[2]Art. I, Liability Convention.

[3] *See,* Foster, *The Convention on International Liability for Damage Caused by Space Objects,* The Can. Y. B. of Int'l. L. 137, 145, (1972). [Hereinafter Foster] Convention on International Liability for Damage Caused by Space Objects - Analysis and Background Data, Staff Report prepared for use of the Committee on Aeronautical and Space Sciences, U. S. Senate 92nd Congr. 2d Sess. (1972) at 25 [hereinafter Staff Report].

There may also be a problem of establishing ownership. Article II requires that Norka be a launching state, and Article I defines a launching state to be one ". . .which launches or procures the launching of a space object" or one ". . . from whose territory or facility a space object is launched." Nat should encounter no substantial difficulty in establishing that Norka is a launching state under most circumstances.[4] The intention of the drafters in the definition of "launching state" may well have been to impose liability upon a broad class of states responsible for putting space objects into orbit that later fall and injure people, but the inclusion of ". . . *its* space object" in the Article[5] which actually imposes that liability puts a cloud upon the effectiveness of the language to carry out that intention. The inclusion of that unfortunate three-letter word of inconsistency may provide fertile ground for contention in a contested case.

My background as a teacher of the common law regarding injuries to persons, property, and relations makes the temptation to deal with the issue of causation for the remainder of my alloted space nearly irresistible. But there are other issues to be raised, so I must be content here to suggest that the Liability Convention is seriously deficient in its failure to elaborate upon such problems as:

(1) does "cause" mean *direct* cause, or does it refer to one of a series of elements in a sequential claim of factual events?

(2) does "cause" mean that the instrumentality contributed to the injury of Nat, or does it mean that the instrumentality caused the event which ultimately resulted in Nat's injury?

(3) can Nat establish "cause" simply by showing he was in proximity to the space object when his injuries were incurred, or must he show that he sustained some physical impact with the object?

(4) will Norka escape if some "greater cause" was involved; does Nat have to negate such a possibility in his claim, or can he require Norka to establish the "greater cause" as a matter of affirmative defense?[6]

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[4] *But see,* Foster, *supra,* at 163-168.

[5] Art. II, Liability Convention.

[6] This question is at least partially answered by Article VI which would allow Norka to be exonerated from absolute liability if it ". . . establishes that the damage has resulted either wholly or partially from gross negligence or from an act or omission done with intent to cause damage on the part of a claimant state or of natural or juridical persons it represents." Query: does it exonerate from liability altogether, or simply from *absolute* liability; does it exonerate from liability to Nat in the event some other person represented by the U. S. besides Nat has engaged in the gross negligence or intentional conduct?

Hoping to have established that the liability of Norka, while *potentially* absolute and unlimited, is far from clear-cut, our attention may turn now to Nat's problems when he is injured by a U.S. space object.

Liability of the U.S. Government:

Since the Liability Convention excludes its coverage from situations where the damage is caused by a launching state to one of its nationals,[7] if Nat is injured by a fallen U.S. space object he must seek redress through other mediums. And, since the tradition of governmental immunity under our system of justice maintains that the government may not be sued without its consent, those mediums for recovery are carefully delineated. If Nat is equally as careful in his attempt to establish that his claim conforms to those delineations, he may obtain recovery.

The Federal government has given its consent to be sued for ". . . money damages. . .for injury or loss of property, or personal injury or death caused by the negligent or wrongful act or omission of any employee. . .".[8] A serious difference between Nat's position vis-a-vis his own government and that with respect to Norka is immediately made apparent; Nat has to prove fault to recover here. The words of the statute "negligent or wrongful act" have been interpreted to mean that liability will be imposed only upon the theory that the employees of the government have failed to exercise ordinary care or have engaged in some other conduct constituting misfeasance or malfeasance.[9] The magnitude of this hurdle in a space object accident is superlative. Nat will have to focus upon specific conduct, in the myriad of human activity that is involved with space operations, and establish that it was sufficiently faulty to make the U.S. government answerable. That task is by no means made easier by the fact that much of the human activity involved and the documentation of it are controlled by tight security measures, limiting Nat's access for discovery purposes. Internal procedures are so well developed that the government is well on the way from the outset toward rebutting Nat's claim that the employees failed to exercise ordinary care. Whatever strength Nat's case may have in sympathies for "the injured citizen against the cold, insensitive government" is lost because the trial of such cases is without a jury.

Even supposing that Nat has successfully overcome the obstacles in establishing negligence, his claim may still fail under the statute if the negligence is deemed to have been the exercise of a *discretionary* function of the employee that committed that act of

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[7]Art. VII, Liability Convention.

[8]28 U.S.C.A. § 1346(b), the Federal Tort Claims Act.

[9]See, *Dalehite v. United States,* 346 U.S. 15 (1953); *Laird v. Nelms,* 406 U.S. 797 (1972). This fault requirement is applicable to the administrative remedy that Nat must pursue prior to actual litigation. Section 2675, requires that he submit his claim to the appropriate agency for determination, and that such a claim be denied by that agency before suit can be brought. Section 2672 places a $25,000 ceiling upon awards to be made under this procedure, without prior approval of the attorney general.

negligence.[10] The closer the act that Nat claims to have been the operative negligent act causing his injury comes to being a policy decision, as opposed to a decision involving the operation of the activity pursuant to that policy, the greater the possibility that it will be considered by the court to be a discretionary function, barring Nat's recovery.[11]

Analysis of the decisions dealing with this exclusion from liability teaches that, while generalizations such as that just stated may assist in predicting the outcome of Nat's claim on the basis of what conduct he asserts to be negligent conduct, precision is far from at hand when it comes down to the actual determination of whether the act is nonactionable under the discretionary exclusion. For example, if Nat were to claim that the failure to warn him prior to the accident was the *sine qua non* of his injury, the application of the exclusion may well turn upon the persuasiveness of the argument concerning the existence of an obligation to warn. The facts of the particular case under review may be sufficiently persuasive to impose a "duty to communicate. . . adequate warning of existing foreseeable hazard," the failure of which to exercise might constitute actionable "operational mismanagement."[12] On the other hand, the failure to warn may be determined to be a "discretionary decision. . . that such notice was impractical or would interfere with the carrying out of the project or would involve wasted time without justification."[13]

Nat's claim will be fraught with the same difficulties if he tries to establish faulty conduct involving the very decision to engage in the project. The sonic boom cases provide illustrations of court decisions that have gone both ways on the question of whether such projects involved discretionary decisions.[14]

Nat has other options under federal law, however, in seeking redress for his injuries in order to avoid the difficulties presented by the Federal Tort Claims Act and its limiting provisions. Where the injuries have arisen from the ". . . conduct of the [National Aeronautics and Space] Administration's functions. . .",[15] Nat may submit

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[10] 28 U.S.C.A. § 2680 (a).

[11] This is a gross oversimplification of an extremely complex and troublesome aspect of the Federal Torts Claims Act, but it must suffice for the purposes of this paper. For more detailed discussion of the problems created for persons in Nat's position, *see,* 2 L. Jayson, Handling Federal Tort Claims: Administrative and Judicial Remedies (Chp. 12 1978) [hereinafter Jayson].

[12] *United States v. State of Washington,* 351 F. 2d 914, 91 (9th Cir., 1965) where decedents were killed when their plane flew into power lines across a valley not marked by warning devices. *Cited* in Jayson, *supra,* note 10, at 12-83 n. 12-1.

[13] *Bulloch v. United States,* 133 F. Supp. 885, 888, 889 (Utah, 1955), where claimants' sheep were allegedly killed by radiation from nuclear testing. *Cited* in Jayson, *supra,* note 10, at 12-84 n. 13.

[14] *See,* Jayson, *supra* note 10, at 12-120-12-124.

[15] 42 U.S.C.A. § 2473 (b) (13) (A).

his claim for personal or property injury, or death, to NASA for special administrative determination and settlement by that agency.[16]

Under this administrative claims procedure, Nat can avoid the requirement of proving that the injury was caused by the fault of federal employees. So long as he establishes that the injury arose out of the conduct of the agency's space programs, he can avail himself of the theory of strict liability. Claims procedure, evidentiary standards, scope of review, and investigatory procedures are spelled out in some detail by the agency's regulations.[17]

The administrative scheme of recovery is not without its drawbacks, however. The power of the agency to pay claims is limited to $5,000.00.[18] In the event of serious injury exceeding this amount, Nat would be forced to undergo the rigors of the requirements of the Federal Tort Claims Act. The agency may mitigate this effect if it decides Nat's claim is meritorious by submitting the claim to Congress for its consideration.[19]

The consideration of Congress just mentioned is the remaining scheme of recovery for Nat under the circumstances of injury by a U.S. space object. It becomes operable, however, only after Nat has exhausted his other available judicial and administrative remedies. By private bill, the Congress has unlimited power to provide redress for Nat's injuries. For example, it may simply provide for payment of a definite sum, it may remove monetary limits from the agency's power to determine and settle Nat's claim, or shape relief in some other form.[20]

Liability of the Private U.S. Corporation:

The situation involving injury by a space object owned by a private U.S. corporation presents further complexities to Nat's attempt to obtain redress. The Liability Convention does not address itself to the accountability of such entities to persons injured by fallen space objects. This is a serious omission in light of the fact that the Outer Space Treaty places responsibility upon parties to that treaty for space activities whether the acting entity is governmental or nongovernmental,[21] and in light

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[16]This administrative procedure is independent of the claims procedure required by the Tort Claims Act, § 28 U.S.C.A. 2675. The claimant may select from the two statutes as alternatives. Failure to designate the statute under which the claimant is submitting a claim allows the agency to treat the matter under either. *See,* Jayson, *supra,* note 10, at 1-90.

[17]14 C.F.R. §§ 1204.900-.915, covering claims accruing on or after Jan. 18, 1967.

[18]42 U.S.C.A. § 2473 (b) (13) (A).

[19]42 U.S.C.A. § 2473 (b) (13) (B).

[20] *See,* Jayson, *supra,* note 10, at 1-164.

[21]Art. VI, Outer Space Treaty.

of the avowed purpose of the Liability Convention to "elaborate effective international rules" on the matter. Arguably, Nat could impose the responsibility of the Outer Space Treaty through the medium of the Liability Convention upon a launching state if the launching state was a party to both documents (and if the launching state was some state other than the U.S.). But what if the launching state was not party to one or both of the agreements?[22] Furthermore, even where the launching state is party to both, the problem of the language ". . . *its* space object" in Article II of the Liability Conventions crops up again.

If Nat decided to proceed against the corporation instead of the launching states on a theory of negligence, he faces the very difficult task of establishing a *prima facie* case. As mentioned in previous discussion, penetrating the extremely complex technical features of the defendant's space activities to identify the culpable conduct with specificity would be difficult, time-consuming, and require the services of people qualified to make the investigation, all of which would make the preparatory work quite expensive. A much better theory to proceed upon would be strict liability.

There seems to be considerable agreement that space activities are of the type of high-risk activities upon which a foundation for the imposition of strict liability can be grounded,[23] but the matter cannot be treated as a foregone conclusion at litigation where the matter would be one of first impression. Again, constraints require brevity and over-simplification, but in addition to having to prove damage and causation as in all the other schemes for recovery, Nat would carry the burden of persuading the court and jury that the activity is one in which the imposition of strict liability is justified. The American synthesis of this principle, enunciated in the leading case of *Rylands v. Fletcher* [24] over one hundred years ago, is contained in the Restatement of Torts 2nd, Sections 519 and 520. Those sections, representing a distillation of the American case law impose strict liability where the activity is determined to be "abnormally dangerous." A court considering Nat's claim, and following the Restatement formulation, would make the determination of whether the defendant's space activities were "abnormally dangerous" by reference to several criteria:

(a) Whether the activity involves a high degree of risk of some harm to the person, land or chattels of others;

(b) Whether the gravity of the harm which may result from it is likely to be great;

(c) Whether the risk cannot be eliminated by the exercise of reasonable care;

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[22] *See,* Foster, *supra,* note 2, at 164.

[23] See Foster, *supra,* note 2, at 150-154; Staff Report, *supra,* note 2, at 26.

[24] *Fletcher v. Rylands,* L.R. 1 Exch. 265 (1866); *Rylands v. Fletcher,* L.R. 3 H.L. 330 (1868).

(d) Whether the activity is not a matter of common usage;

(e) Whether the activity is inappropriate to the place where it is carried on; and

(f) The value of the activity to the community.

In evaluating those criteria, it is apparent that each involves the application of value judgments regarding the defendant's activities. In reaching the ultimate conclusion the weight to be given each of the criteria will vary according to the strength of the value judgment accorded to each. From Nat's point of view, criteria (d) through (f) are troublesome. It is not at all apparent that a finder of fact would not attach overriding weight to the importance of space activities and require that Nat bear the risk of injury.

Even assuming that Nat would be successful in imposing a strict liability standard against the corporation, he still faces some extreme difficulty in establishing cause. The imposition of strict liability is nowhere considered to be simply a matter of making the defendant the insurers for plaintiffs' injuries. Because of the involvement of the launching state in actually putting the space object into orbit, the causative factors are multiplied and complicated. The more causative factors present, the greater the number of opportunities for the corporation to escape accountability.

As a final illustrative hypothetical, consider the problems that might be presented if all three of our different types of space objects were involved in the same accident. Suppose that through some negligent act, the private corporation caused the orbit of its space object to rapidly decay. Suppose further that the U.S., after obtaining operational status with the Space Shuttle program, attempted to "rescue" the private space object by bringing it back to earth. As the Space Shuttle began its descent, it encountered the space object of Norka. Recognizing that his reentry attitude would be jeopardized if he made adjustments to avoid the Norka space object, the commander of the Space Shuttle decided to take his chances of missing the Norka craft, since Space Shuttle is many times the size of the former. A collision occurred, the force of which was not anticipated by the commander. The impact causes: (a) the Norka space object to fall from orbit; (b) the corporation's space object to be jarred loose from the space Shuttle and fall from orbit, and; (c) the Space Shuttle to fall uncontrollably to the surface of the earth. All three objects' crashes produce "damage" to Nat the sense contemplated by the Liability Convention.

With all of the substantive bases for recovery available, and three separate respondents available, it would seem that Nat stands a very good chance for full recovery. A closer look reveals otherwise.

First of all, Nat's claim against Norka will be vigorously defended on the basis that its space object did not "cause" Nat's damage. Norka's representatives will argue that "caused by its space object" calls for application of the *sine qua non test:* but for the

acts of the Space Shuttle commander and the corporation's employees, Norka's space object would have remained in orbit. They may also argue that Article VI operates to exonerate them on the grounds that the Space Shuttle commander's decision amounted to gross negligence, since it constituted a reckless disregard for the property and passage rights of Norka.

As to the claim against NASA, Nat will have to prove fault as to any damages in excess of $5,000.00. Even if he is successful in that regard, it is still open to question whether the decision of the Space Shuttle commander was sufficiently discretionary to exonerate NASA from liability under the Federal Tort Claims Act.

IF Nat proceeds against the U.S. corporation, he faces the difficulties of proving negligence, or convincing the court to apply a strict liability standard. Even if successful in either respect, the corporation will, of course, argue vigorously, and with merit, that the acts of the Space Shuttle commander constituted an intervening causative factor which should absolve the corporation from liability.

So, it appears that considerable doubt remains as to the efficacy of each of the regimes discussed in protecting persons injured as a result of fallen space objects. In light of the very real possibility of future accidents brought to light in the wake of the recent Cosmos incident, attention must be given internationally and domestically to insure that it will not be the innocent injured person that bears the risk of such injuries.